# CS229 Section: Linear Algebra

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Slides adapted from past CS229 teams

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# Outline

1. Basic Concepts and Notation
2. Matrix Multiplication
3. Operations and Properties
4. Matrix Calculus

# Basic Concepts and Notation

# Basic Notation

- By $x \in \mathbb{R}^n$, we denote a vector with $n$ entries.

$$\underset{(n \times 1)}{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$$

- By $A \in \mathbb{R}^{m \times n}$ we denote a matrix with $m$ rows and $n$ columns, where the entries of $A$ are real numbers.

$$\underset{(m \times n)}{A} = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix} = \underset{(m \times 1) \longrightarrow n}{\begin{bmatrix} | & | & & | \\ a^1 & a^2 & \cdots & a^n \\ | & | & & | \end{bmatrix}} = \begin{bmatrix} — & a_1^T & — \\ — & a_2^T & — \\ & \vdots & \\ — & a_m^T & — \end{bmatrix} \begin{array}{l} (1 \times n) \\ \downarrow \\ m \end{array}.$$

# The Identity Matrix

The ***identity matrix***, denoted $I \in \mathbb{R}^{n \times n}$, is a square matrix with ones on the diagonal and zeros everywhere else. That is,

$$I_{ij} = \begin{cases} 1 & i = j \\ 0 & i \neq j \end{cases}$$

It has the property that for all $A \in \mathbb{R}^{m \times n}$,

$$\boxed{AI = A = IA.}$$

$$\begin{bmatrix} 1 & \cdots & 0 \\ \vdots & 1 & \vdots \\ 0 & \cdots & 1 \end{bmatrix}$$

$(n \times n)$

# Diagonal matrices

A ***diagonal matrix*** is a matrix where all non-diagonal elements are 0. This is typically denoted $D = \mathrm{diag}(d_1, d_2, \ldots, d_n)$, with

$$D_{ij} = \begin{cases} d_i & i = j \\ 0 & i \neq j \end{cases}$$

$$\begin{bmatrix} d_1 & & & 0 \\ & d_2 & & \\ & & \ddots & \\ 0 & & & d_n \end{bmatrix}$$

Clearly, $I = \mathrm{diag}(1, 1, \ldots, 1)$.

# Outline


1 Basic Concepts and Notation

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3 Operations and Properties

4 Matrix Calculus

# Vector-Vector Product

- *inner product* or *dot product*

vector vector

$$\underbrace{x}_{\text{vector}}{}^T \underbrace{y}_{\text{vector}} \in \mathbb{R} = \underbrace{\begin{bmatrix} x_1 & x_2 & \cdots & x_n \end{bmatrix}}_{(1 \times n)} \underbrace{\begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}}_{(n \times 1)} = \underbrace{\sum_{i=1}^{n} x_i y_i}_{(1 \times 1)}.$$

equal

scalar

- *outer product*

(m × 1) need not be equal (1 × n)

vector matrix

$$xy^T \in \mathbb{R}^{m\times n} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_m \end{bmatrix} \begin{bmatrix} y_1 & y_2 & \cdots & y_n \end{bmatrix} = \begin{bmatrix} x_1y_1 & x_1y_2 & \cdots & x_1y_n \\ x_2y_1 & x_2y_2 & \cdots & x_2y_n \\ \vdots & \vdots & \ddots & \vdots \\ x_my_1 & x_my_2 & \cdots & x_my_n \end{bmatrix}.$$

vector

(m × n)

matrix

## Matrix-Vector Product

connect to class notes $h(x^{(i)}) = (x^{(i)})^T \theta$, $h(X) = X\theta$

- If we write $A$ by rows, then we can express $Ax$ as,

$(1 \times n)$ $(n \times 1)$ $(1 \times 1)$

$$y = Ax = \begin{bmatrix} — & a_1^T & — \\ — & a_2^T & — \\ & \vdots & \\ — & a_m^T & — \end{bmatrix} x = \begin{bmatrix} a_1^T x \\ a_2^T x \\ \vdots \\ a_m^T x \end{bmatrix}.$$

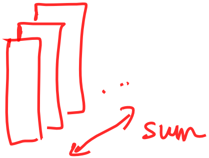
vector of inner products

# Matrix-Vector Product

- If we write $A$ by columns, then we have:

$$y = Ax = \overset{(m \times n)}{\begin{bmatrix} | & | & & | \\ a^1 & a^2 & \cdots & a^n \\ | & | & & | \end{bmatrix}}_{(m\times 1)} \overset{(n \times 1)}{\begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}} = \underbrace{\begin{bmatrix} a^1 \end{bmatrix}}_{(m\times 1)} x_1 + \underbrace{\begin{bmatrix} a^2 \end{bmatrix}}_{(m\times 1)} x_2 + \ldots + \begin{bmatrix} a^n \end{bmatrix} x_n \ . \tag{1}$$

$y$ is a ***linear combination*** of the *columns* of $A$.



# Matrix-Vector Product

It is also possible to multiply on the left by a row vector.

- If we write $A$ by columns, then we can express $x^\top A$ as,

(m×n)

$$y^T = x^T A = x^T \begin{bmatrix} | & | & & | \\ a^1 & a^2 & \cdots & a^n \\ | & | & & | \end{bmatrix} = \begin{bmatrix} x^T a^1 & x^T a^2 & \cdots & x^T a^n \end{bmatrix}$$

(1×m) (m×1) (1×1) $\longrightarrow$ n

(1×n)

# Matrix-Vector Product

It is also possible to multiply on the left by a row vector.

- expressing $A$ in terms of rows we have:

$(m \times n)$

$$y^T = x^T A = \underbrace{\begin{bmatrix} x_1 & x_2 & \cdots & x_m \end{bmatrix}}_{(1 \times m)} \begin{bmatrix} — & a_1^T & — \\ — & a_2^T & — \\ & \vdots & \\ — & a_m^T & — \end{bmatrix} \; (1 \times n)$$

$$= x_1 \underbrace{\begin{bmatrix} — & a_1^T & — \end{bmatrix}}_{(1 \times n)} + x_2 \begin{bmatrix} — & a_2^T & — \end{bmatrix} + \ldots + x_m \begin{bmatrix} — & a_m^T & — \end{bmatrix}$$

$y^T$ is a linear combination of the *rows* of $A$.

sum

# Matrix-Matrix Multiplication (different views)

1. As a set of vector-vector products (dot product)

$$\underset{(m\times p)}{C} = AB = \underset{(m\times n)}{\begin{bmatrix} — & a_1^T & — \\ — & a_2^T & — \\ & \vdots \; i & \\ — & a_m^T & — \end{bmatrix}} \underset{(n\times p)}{\begin{bmatrix} | & | & & | \\ b^1 & b^2 & \cdots \; j & b^p \\ | & | & & | \end{bmatrix}} = \begin{bmatrix} a_1^T b^1 & a_1^T b^2 & \cdots & a_1^T b^p \\ a_2^T b^1 & a_2^T b^2 & \cdots & a_2^T b^p \\ \vdots & \vdots & \ddots \; ij & \vdots \\ a_m^T b^1 & a_m^T b^2 & \cdots & a_m^T b^p \end{bmatrix}.$$

$$\underset{(1\times 1)}{c_{ij}} = \underset{(1\times n)}{a_i^T} \; \underset{(n\times 1)}{b^j}$$

# Matrix-Matrix Multiplication (different views)

2. As a sum of outer products

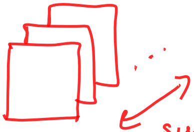
(m × p) (p × n)

$$C = AB = \begin{bmatrix} | & | & & | \\ a^1 & a^2 & \cdots & a^p \\ | & | & & | \end{bmatrix} \begin{bmatrix} — & b_1^T & — \\ — & b_2^T & — \\ & \vdots & \\ — & b_p^T & — \end{bmatrix} = \sum_{i=1}^{p} a^i b_i^T .$$

(m × n)

(m×1) → p

(1× n)



sum

# Matrix-Matrix Multiplication (different views)

3. As a set of matrix-vector products.

$$C = AB = A \begin{bmatrix} | & | & & | \\ b^1 & b^2 & \cdots & b^n \\ | & | & & | \end{bmatrix} = \begin{bmatrix} | & | & & | \\ Ab^1 & Ab^2 & \cdots & Ab^n \\ | & | & & | \end{bmatrix}. \tag{2}$$

Here the $i$th column of $C$ is given by the matrix-vector product with the vector on the right, $c_i = Ab_i$. These matrix-vector products can in turn be interpreted using both viewpoints given in the previous subsection.

## Matrix-Matrix Multiplication (different views)

4. As a set of vector-matrix products.

$$C = AB = \begin{bmatrix} — & a_1^T & — \\ — & a_2^T & — \\ & \vdots & \\ — & a_m^T & — \end{bmatrix} B = \begin{bmatrix} — & a_1^T B & — \\ — & a_2^T B & — \\ & \vdots & \\ — & a_m^T B & — \end{bmatrix}.$$

$j^{th}$ row of $C$ : $a_j^T B$

# Matrix-Matrix Multiplication (properties)

$\neq BAC$ (order is X)

- Associative: $(AB)C = A(BC)$.
- Distributive: $A(B + C) = AB + AC$.

prove!

not necessary

- In general, *not* commutative; that is, it can be the case that $\boxed{AB \neq BA}$. (For example, if $A \in \mathbb{R}^{m\times n}$ and $B \in \mathbb{R}^{n\times q}$, the matrix product $BA$ does not even exist if $m$ and $q$ are not equal!)

counter example!

# Outline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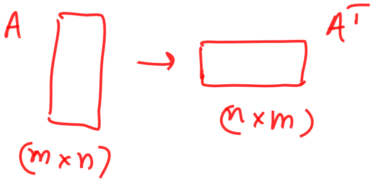
# Operations and Properties

# The Transpose

The ***transpose*** of a matrix results from "flipping" the rows and columns. Given a matrix $A \in \mathbb{R}^{m\times n}$, its transpose, written $A^T \in \mathbb{R}^{n\times m}$, is the $n \times m$ matrix whose entries are given by

$$(A^T)_{ij} = A_{ji}.$$




The following properties of transposes are easily verified:

- $(A^T)^T = A$ → flip twice
- $(AB)^T = \underline{B^T}\,\underline{A^T}$ order changes!
- $(A+B)^T = A^T + B^T$

} prove!

sum → flip ≡ flip → sum

# Trace

The ***trace*** of a square matrix $A \in \mathbb{R}^{n\times n}$, denoted $\mathrm{tr}A$, is the sum of diagonal elements in the matrix:

$$\mathrm{tr}A = \sum_{i=1}^{n} A_{ii}.$$

$$\begin{bmatrix} a_{11} & & & \\ & a_{22} & & \\ & & \ddots & \\ & & & a_{nn} \end{bmatrix}$$

The trace has the following properties:

- For $A \in \mathbb{R}^{n\times n}$, $\mathrm{tr}A = \mathrm{tr}A^T$. → flipping doesn't change diagonals
- For $A, B \in \mathbb{R}^{n\times n}$, $\mathrm{tr}(A + B) = \mathrm{tr}A + \mathrm{tr}B$.
- For $A \in \mathbb{R}^{n\times n}$, $t \in \mathbb{R}$, $\mathrm{tr}(tA) = t\,\mathrm{tr}A$. verify!
- For $A, B$ such that $AB$ is square, $\boxed{\mathrm{tr}AB = \mathrm{tr}BA.}$ diag. are special!
- For $A, B, C$ such that $ABC$ is square, $\mathrm{tr}ABC = \mathrm{tr}BCA = \mathrm{tr}CAB$, and so on for the product of more matrices. Prove!

$\neq \mathrm{tr}\,ACB$ (order is $\neq$.)

# Norms

A ***norm*** of a vector $\|x\|$ is informally a measure of the "length" of the vector.

More formally, a norm is any function $f : \mathbb{R}^n \to \mathbb{R}$ that satisfies 4 properties:

1. For all $x \in \mathbb{R}^n$, $f(x) \geq 0$ (non-negativity).
2. $f(x) = 0$ if and only if $x = 0$ (definiteness). length !
3. For all $x \in \mathbb{R}^n$, $t \in \mathbb{R}$, $f(tx) = |t| f(x)$ (homogeneity). scaling comp. scales length
4. For all $x, y \in \mathbb{R}^n$, $f(x + y) \leq f(x) + f(y)$ (triangle inequality). (geometry)

# Examples of Norms

The commonly-used Euclidean or $\ell_2$ norm,

$$\|x\|_2 = \sqrt{\sum_{i=1}^{n} x_i^2}.$$

$$x = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$$

The $\ell_1$ norm,

$$\|x\|_1 = \sum_{i=1}^{n} |x_i|$$

The $\ell_\infty$ norm,

$$\|x\|_\infty = \max_i |x_i|.$$

# Examples of Norms

In fact, all three norms presented so far are examples of the family of $\ell_p$ norms, which are parameterized by a real number $p \geq 1$, and defined as

$$\boxed{\|x\|_p = \left(\sum_{i=1}^{n} |x_i|^p\right)^{1/p}.}$$

# Matrix Norms

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ \vdots & \ddots & & \vdots \\ a_{1m} & \cdots & & a_{mn} \end{bmatrix}$$

Norms can also be defined for matrices, such as the Frobenius norm,

$$\|A\|_F = \sqrt{\sum_{i=1}^{m} \sum_{j=1}^{n} A_{ij}^2} = \sqrt{\operatorname{tr}(A^T A)}.$$

useful, will see this again & again

Many other norms exist, but they are beyond the scope of this review.

# Linear Independence

A set of vectors $\{x_1, x_2, \dots x_n\} \subset \mathbb{R}^m$ is said to be ***(linearly) dependent*** if one vector belonging to the set *can* be represented as a linear combination of the remaining vectors; that is, if

any

$$x_n = \sum_{i=1}^{n-1} \alpha_i x_i$$

Scalar coeff.

for some scalar values $\alpha_1, \dots, \alpha_{n-1} \in \mathbb{R}$; otherwise, the vectors are ***(linearly) independent***.

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**Example**:

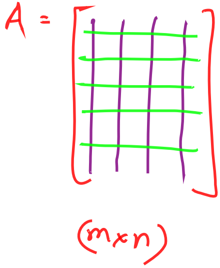
$$\overset{-2}{x_1 = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}} \quad \overset{+1}{x_2 = \begin{bmatrix} 4 \\ 1 \\ 5 \end{bmatrix}} \quad x_3 = \begin{bmatrix} 2 \\ -3 \\ -1 \end{bmatrix}$$

are linearly dependent because $\underline{x_3 = -2x_1 + x_2}$.

# Rank of a Matrix

- The ***column rank*** of a matrix $A \in \mathbb{R}^{m \times n}$ is the largest number of columns of $A$ that constitute a linearly independent set.

col. rank $\leq$ # cols $= n$



# Rank of a Matrix

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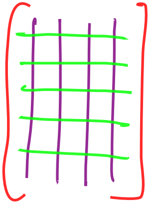
- The ***row rank*** is the largest number of rows of $A$ that constitute a linearly independent set.

row rank $\leq$ # rows $= m$

$A =$  $(m \times n)$

# Rank of a Matrix

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- The ***row rank*** is the largest number of rows of $A$ that constitute a linearly independent set.
- For any matrix $A \in \mathbb{R}^{m \times n}$, it turns out that the column rank of $A$ is equal to the row rank of $A$ (prove it yourself!), and so both quantities are referred to collectively as the ***rank*** of $A$, denoted as $\mathrm{rank}(A)$.

(prove!)

col. rank of A = row rank of A = rank(A)

# Properties of the Rank

- For $A \in \mathbb{R}^{m \times n}$, $\mathrm{rank}(A) \leq \min(m, n)$. If $\boxed{\mathrm{rank}(A) = \min(m, n)}$, then $A$ is said to be ***full rank***.

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- For $A \in \mathbb{R}^{m \times n}$, $\mathrm{rank}(A) = \mathrm{rank}(A^T)$. flipping doesn't change linear independence!

## Properties of the Rank

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- For $A \in \mathbb{R}^{m \times n}$, $\mathrm{rank}(A) = \mathrm{rank}(A^T)$.
- For $A \in \mathbb{R}^{m \times p}$, $B \in \mathbb{R}^{p \times n}$, $\mathrm{rank}(AB) \leq \min(\mathrm{rank}(A), \mathrm{rank}(B))$.

# Properties of the Rank

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- For $A \in \mathbb{R}^{m \times n}$, $\mathrm{rank}(A) = \mathrm{rank}(A^T)$.
- For $A \in \mathbb{R}^{m \times p}$, $B \in \mathbb{R}^{p \times n}$, $\mathrm{rank}(AB) \leq \min(\mathrm{rank}(A), \mathrm{rank}(B))$.
- For $A, B \in \mathbb{R}^{m \times n}$, $\mathrm{rank}(A + B) \leq \mathrm{rank}(A) + \mathrm{rank}(B)$.

# The Inverse of a Square Matrix

- The ***inverse*** of a square matrix $A \in \mathbb{R}^{n \times n}$ is denoted $\boxed{A^{-1}}$, and is the unique matrix such that
$$\boxed{A^{-1}A = I = AA^{-1}.}$$

Does this always exist?

# The Inverse of a Square Matrix

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$$A^{-1}A = I = AA^{-1}.$$

- We say that $A$ is ***invertible*** or ***non-singular*** if $A^{-1}$ exists and ***non-invertible*** or ***singular*** otherwise.

When does it exist ??

# The Inverse of a Square Matrix

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- We say that $A$ is ***invertible*** or ***non-singular*** if $A^{-1}$ exists and ***non-invertible*** or ***singular*** otherwise.

- In order for a square matrix $A$ to have an inverse $A^{-1}$, then $\boxed{A \text{ must be full rank.}}$

if $A \rightarrow$ full rank : invertible
square non-singular

# The Inverse of a Square Matrix

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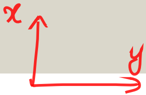
$$A^{-1}A = I = AA^{-1}.$$
- We say that $A$ is ***invertible*** or ***non-singular*** if $A^{-1}$ exists and ***non-invertible*** or ***singular*** otherwise.
- In order for a square matrix $A$ to have an inverse $A^{-1}$, then $A$ must be full rank.
- Properties (Assuming $A, B \in \mathbb{R}^{n\times n}$ are non-singular):
  - ✓ $(A^{-1})^{-1} = A$
  - $(AB)^{-1} = \underline{B^{-1}}\,\underline{A^{-1}}$ order!
  - $(A^{-1})^T = (A^T)^{-1}$. For this reason this matrix is often denoted $A^{-T}$.

prove!

# Orthogonal Matrices




- Two vectors $x, y \in \mathbb{R}^n$ are ***orthogonal*** if $x^T y = 0$.
- A vector $x \in \mathbb{R}^n$ is ***normalized*** if $\|x\|_2 = 1$.
- A square matrix $U \in \mathbb{R}^{n \times n}$ is ***orthogonal*** if all its columns are orthogonal to each other and are normalized (the columns are then referred to as being ***orthonormal***).

$$U = \begin{bmatrix} | & | & & | \\ u_1 & u_2 & \cdots & u_n \\ | & | & & | \end{bmatrix} \quad (n \times n)$$

cols: orthogonal to each other
normalized

$u_i^T u_j = \boxed{0}$ $(j \neq i)$ $\qquad u_i^T u_i = \boxed{1}$

## Orthogonal Matrices

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- **Properties:**
  - The inverse of an orthogonal matrix is its transpose.

$$\boxed{U^T U = I = U U^T.}$$

# Orthogonal Matrices

- Two vectors $x, y \in \mathbb{R}^n$ are ***orthogonal*** if $x^T y = 0$.
- A vector $x \in \mathbb{R}^n$ is ***normalized*** if $\|x\|_2 = 1$.
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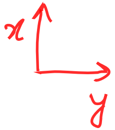
- **Properties:**
  - The inverse of an orthogonal matrix is its transpose.
  $$U^T U = I = UU^T.$$
  - Operating on a vector with an orthogonal matrix will not change its Euclidean norm, i.e.,
  $$\boxed{\|Ux\|_2 = \|x\|_2}$$
  no scaling in $l_2$ norm

  for any $x \in \mathbb{R}^n$, $U \in \mathbb{R}^{n \times n}$ orthogonal.

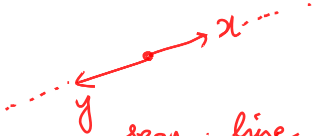
## Span and Projection

- The *span* of a set of vectors $\{x_1, x_2, \ldots x_n\}$ is the set of all vectors that can be expressed as a linear combination of $\{x_1, \ldots, x_n\}$. That is,

$$\operatorname{span}(\{x_1, \ldots x_n\}) = \left\{ v : v = \sum_{i=1}^{n} \alpha_i x_i, \quad \alpha_i \in \mathbb{R} \right\}.$$




span : $\mathbb{R}^2$ plane




span : line

## Span and Projection

- The ***span*** of a set of vectors $\{x_1, x_2, \ldots x_n\}$ is the set of all vectors that can be expressed as a linear combination of $\{x_1, \ldots, x_n\}$. That is,

$$\operatorname{span}(\{x_1, \ldots x_n\}) = \left\{ v : v = \sum_{i=1}^{n} \alpha_i x_i, \quad \alpha_i \in \mathbb{R} \right\}.$$

- The ***projection*** of a vector $y \in \mathbb{R}^m$ onto the span of $\{x_1, \ldots, x_n\}$ is the vector $v \in \operatorname{span}(\{x_1, \ldots x_n\})$, such that $v$ is as close as possible to $y$, as measured by the Euclidean norm $\|v - y\|_2$.

$$\operatorname{Proj}(y; \{x_1, \ldots x_n\}) = \boxed{\operatorname{argmin}_{v \in \operatorname{span}(\{x_1, \ldots, x_n\})} \|y - v\|_2.}$$

# Range

- The ***range*** or the column space of a matrix $A \in \mathbb{R}^{m \times n}$, denoted $\mathcal{R}(A)$, is the the span of the columns of $A$. In other words,

$$\mathcal{R}(A) = \{v \in \mathbb{R}^m : \boxed{v = Ax,} x \in \mathbb{R}^n\}.$$

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$$\mathcal{R}(A) = \{v \in \mathbb{R}^m : v = Ax, x \in \mathbb{R}^n\}.$$

- Assuming $A$ is full rank and $n < m$, the projection of a vector $y \in \mathbb{R}^m$ onto the range of $A$ is given by,

$$\mathrm{Proj}(y; A) = \boxed{\mathrm{argmin}_{v \in \mathcal{R}(A)} \|v - y\|_2.}$$

# Null space

The ***nullspace*** of a matrix $A \in \mathbb{R}^{m \times n}$, denoted $\mathcal{N}(A)$ is the set of all vectors that equal 0 when multiplied by $A$, i.e.,

$$\mathcal{N}(A) = \{x \in \mathbb{R}^n : Ax = 0\}.$$

remember orthogonal vectors !

(Rank-nullity theorem)

# The Determinant

matrix    scalar

The ***determinant*** of a square matrix $A \in \mathbb{R}^{n \times n}$, is a function $\det : \mathbb{R}^{n \times n} \to \mathbb{R}$, and is denoted $|A|$ or $\det A$.
Given a matrix

$$\begin{bmatrix} — & a_1^T & — \\ — & a_2^T & — \\ & \vdots & \\ — & a_n^T & — \end{bmatrix},$$

geometrically:

consider the set of points $S \subset \mathbb{R}^n$ as follows:

restricted span

$$S = \{v \in \mathbb{R}^n : v = \sum_{i=1}^{n} \alpha_i a_i \text{ where } 0 \leq \alpha_i \leq 1, i = 1, \ldots, n\}.$$

The absolute value of the determinant of $A$ is a measure of the $\boxed{\text{“volume”}}$ of the set $S$.

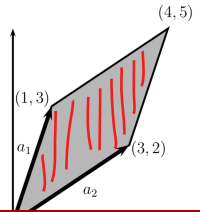
# The Determinant: Intuition

For example, consider the $2 \times 2$ matrix,

$$A = \begin{bmatrix} 1 & 3 \\ 3 & 2 \end{bmatrix} \tag{3}$$

Here, the rows of the matrix are

$$a_1 = \begin{bmatrix} 1 \\ 3 \end{bmatrix} \quad a_2 = \begin{bmatrix} 3 \\ 2 \end{bmatrix}$$




parallelogram

## The Determinant: Properties

Algebraically, the determinant satisfies the following three properties:

1. The determinant of the identity is 1, $|I| = 1$. (Geometrically, the volume of a unit hypercube is 1).

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In case you are wondering, it is not immediately obvious that a function satisfying the above three properties exists. In fact, though, such a function does exist, and is unique (which we will not prove here).

# The Determinant: Properties

- For $A \in \mathbb{R}^{n \times n}$, $|A| = |A^T|$.
- For $A, B \in \mathbb{R}^{n \times n}$, $|AB| = |A||B|$.

not full rank !

- For $A \in \mathbb{R}^{n \times n}$, $|A| = 0$ if and only if $A$ is singular (i.e., non-invertible). (If $A$ is singular then it does not have full rank, and hence its columns are linearly dependent. In this case, the set $S$ corresponds to a "flat sheet" within the $n$-dimensional space and hence has zero volume.)
- For $A \in \mathbb{R}^{n \times n}$ and $A$ non-singular, $|A^{-1}| = 1/|A|$.

## The Determinant: Formula !! $\frac{\prime\,\prime}{0}$

Let $A \in \mathbb{R}^{n\times n}$, $A_{\backslash i,\backslash j} \in \mathbb{R}^{(n-1)\times(n-1)}$ be the *matrix* that results from deleting the $i$th row and $j$th column from $A$.
The general (recursive) formula for the determinant is

$$\begin{aligned} |A| &= \sum_{i=1}^{n} (-1)^{i+j} a_{ij} |A_{\backslash i,\backslash j}| \quad &&(\text{for any } j \in 1,\ldots,n) \\ &= \sum_{j=1}^{n} (-1)^{i+j} a_{ij} |A_{\backslash i,\backslash j}| \quad &&(\text{for any } i \in 1,\ldots,n) \end{aligned}$$

with the initial case that $|A| = a_{11}$ for $A \in \mathbb{R}^{1\times 1}$. If we were to expand this formula completely for $A \in \mathbb{R}^{n\times n}$, there would be a total of $n!$ ($n$ factorial) different terms. For this reason, we hardly ever explicitly write the complete equation of the determinant for matrices bigger than $3 \times 3$.

# The Determinant: Examples

However, the equations for determinants of matrices up to size $3 \times 3$ are fairly common, and it is good to know them:

$$
\begin{aligned}
\left|[a_{11}]\right| &= a_{11} \\
\left|\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}\right| &= a_{11}a_{22} - a_{12}a_{21} \\
\left|\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}\right| &= \begin{array}{c} a_{11}a_{22}a_{33} + a_{12}a_{23}a_{31} + a_{13}a_{21}a_{32} \\ \quad -a_{11}a_{23}a_{32} - a_{12}a_{21}a_{33} - a_{13}a_{22}a_{31} \end{array}
\end{aligned}
$$

# Quadratic Forms

Given a square matrix $A \in \mathbb{R}^{n \times n}$ and a vector $x \in \mathbb{R}^n$, the scalar value $x^T Ax$ is called a ***quadratic form***. Written explicitly, we see that

$$x^T Ax = \sum_{i=1}^{n} x_i (Ax)_i = \sum_{i=1}^{n} x_i \left( \sum_{j=1}^{n} A_{ij} x_j \right) = \sum_{i=1}^{n} \sum_{j=1}^{n} A_{ij} x_i x_j \ .$$

every element of A

every element of x

every element of x

scalar

$\Downarrow$

$(x^T A x)^T = x^T A x$

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We often implicitly assume that the matrices appearing in a quadratic form are only symmetric.

parts contribute to Q. Form

$$\underbrace{x^T Ax}_{\text{Scalar}} = (x^T Ax)^T = x^T A^T x = x^T \left( \frac{1}{2} A + \frac{1}{2} A^T \right) x,$$

# Positive Semidefinite Matrices

A symmetric matrix $A \in \mathbb{S}^n$ is:

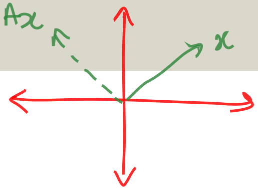
- ***positive definite*** (PD), denoted $A \succ 0$ if for all non-zero vectors $x \in \mathbb{R}^n$, $x^T Ax > 0$.
- ***positive semidefinite*** (PSD), denoted $A \succeq 0$ if for all vectors $x^T Ax \geq 0$.
- ***negative definite*** (ND), denoted $A \prec 0$ if for all non-zero $x \in \mathbb{R}^n$, $x^T Ax < 0$.
- ***negative semidefinite*** (NSD), denoted $A \preceq 0$ ) if for all $x \in \mathbb{R}^n$, $x^T Ax \leq 0$.
- ***indefinite***, if it is neither positive semidefinite nor negative semidefinite — i.e., if there exists $x_1, x_2 \in \mathbb{R}^n$ such that $x_1^T Ax_1 > 0$ and $x_2^T Ax_2 < 0$.

## Positive Semidefinite Matrices

- One important property of positive definite and negative definite matrices is that they are always full rank, and hence, invertible. det $\neq 0$
- Given any matrix $A \in \mathbb{R}^{m \times n}$ (not necessarily symmetric or even square), the matrix $G = A^T A$ (sometimes called a ***Gram matrix***) is always positive semidefinite. Further, if $m \geq n$ and $A$ is full rank, then $G = A^T A$ is positive definite.

any $A \longrightarrow \text{Gram}(A) = A^T A$

# Eigenvalues and Eigenvectors





$Ax \rightarrow$ transformation of $x$ using $A$

Given a square matrix $A \in \mathbb{R}^{n \times n}$, we say that $\lambda \in \boxed{\mathbb{C}}$ is an ***eigenvalue*** of $A$ and $x \in \mathbb{C}^n$ is the corresponding ***eigenvector*** if

$$\boxed{Ax = \lambda x,} \quad x \neq 0.$$

Intuitively, this definition means that multiplying $A$ by the vector $x$ results in a new vector that points in the same direction as $x$, but scaled by a factor $\lambda$.

# Eigenvalues and Eigenvectors

$Ax = \lambda I x$

We can rewrite the equation above to state that $(\lambda, x)$ is an eigenvalue-eigenvector pair of $A$ if,

$$\boxed{(\lambda I - A)x = 0,} \quad x \neq 0.$$

But $(\lambda I - A)x = 0$ has a non-zero solution to $x$ if and only if $(\lambda I - A)$ has a non-empty nullspace, which is only the case if $(\lambda I - A)$ is singular, i.e.,

To find $\lambda$

$$\Rightarrow \quad |(\lambda I - A)| = 0.$$

We can now use the previous definition of the determinant to expand this expression $|(\lambda I - A)|$ into a (very large) polynomial in $\lambda$, where $\lambda$ will have degree $n$. It's often called the **characteristic polynomial** of the matrix $A$.

# Properties of eigenvalues and eigenvectors

- The trace of a $A$ is equal to the sum of its eigenvalues,

$$\mathrm{tr}A = \sum_{i=1}^{n} \lambda_i.$$

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- The rank of $A$ is equal to the number of non-zero eigenvalues of $A$.
- Suppose $A$ is non-singular with eigenvalue $\lambda$ and an associated eigenvector $x$. Then $1/\lambda$ is an eigenvalue of $A^{-1}$ with an associated eigenvector $x$, i.e., $A^{-1}x = (1/\lambda)x$.

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- The eigenvalues of a diagonal matrix $D = \mathrm{diag}(d_1, \ldots d_n)$ are just the diagonal entries $d_1, \ldots d_n$.

# Eigenvalues and Eigenvectors of Symmetric Matrices

Throughout this section, let's assume that $A$ is a symmetric real matrix (i.e., $A^\top = A$). We have the following properties:

1. All eigenvalues of $A$ are real numbers. We denote them by $\lambda_1, \ldots, \lambda_n$.
2. There exists a set of eigenvectors $u_1, \ldots, u_n$ such that (i) for all $i$, $u_i$ is an eigenvector with eigenvalue $\lambda_i$ and (ii) $u_1, \ldots, u_n$ are unit vectors and orthogonal to each other.

orthonormal vectors $u_1, u_2 \ldots, u_n$ for $A$ (symmetric) with $\lambda_1, \lambda_2 \ldots, \lambda_n$ eigen values

# New Representation for Symmetric Matrices

- Let $U$ be the orthonormal matrix that contains $u_i$'s as columns:

$A : n \times n$

$$U = \begin{bmatrix} | & | & & | \\ u_1 & u_2 & \cdots & u_n \\ | & | & & | \end{bmatrix}$$

$(n \times n)$

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$$U = \begin{bmatrix} | & | & & | \\ u_1 & u_2 & \cdots & u_n \\ | & | & & | \end{bmatrix}$$

- Let $\Lambda = \mathrm{diag}(\lambda_1, \ldots, \lambda_n)$ be the diagonal matrix that contains $\lambda_1, \ldots, \lambda_n$.

$$AU = \begin{bmatrix} | & | & & | \\ \boxed{Au_1} & Au_2 & \cdots & Au_n \\ | & | & & | \end{bmatrix} = \begin{bmatrix} | & | & & | \\ \boxed{\lambda_1 u_1} & \lambda_2 u_2 & \cdots & \lambda_n u_n \\ | & | & & | \end{bmatrix} = U\mathrm{diag}(\lambda_1, \ldots, \lambda_n) = U\Lambda$$

$$\boxed{AU = U\Lambda}$$

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- Recalling that orthonormal matrix $U$ satisfies that $UU^T = I$, we can diagonalize matrix $A$:

$$A = AUU^T = U\Lambda U^T \tag{4}$$

# Background: representing vector w.r.t. another basis

- Any orthonormal matrix $U = \begin{bmatrix} | & | & & | \\ u_1 & u_2 & \cdots & u_n \\ | & | & & | \end{bmatrix}$ defines a new basis of $\mathbb{R}^n$.

changing the coordinate axes !

- For any vector $x \in \mathbb{R}^n$ can be represented as a linear combination of $u_1, \ldots, u_n$ with coefficient $\hat{x}_1, \ldots, \hat{x}_n$:

$$x = \hat{x}_1 u_1 + \cdots + \hat{x}_n u_n = U\hat{x}$$

$$\boxed{x = U\hat{x}}$$

- Indeed, such $\hat{x}$ uniquely exists

$$\boxed{x = U\hat{x} \Leftrightarrow U^T x = \hat{x}}$$

$$\Rightarrow \boxed{\hat{x} = U^T x}$$

In other words, the vector $\hat{x} = U^T x$ can serve as another representation of the vector $x$ w.r.t the basis defined by $U$.

# “Diagonalizing” matrix-vector multiplication

- Left-multiplying matrix $A$ can be viewed as left-multiplying a diagonal matrix w.r.t the basic of the eigenvectors.
  - Suppose $x$ is a vector and $\hat{x}$ is its representation w.r.t to the basis of $U$. $\hat{x} = U^T x$
  - Let $z = Ax$ be the matrix-vector product.
  - the representation $z$ w.r.t the basis of $U$:

$$\hat{z} = U^T z = U^T Ax = U^T \underbrace{U}_{I}\Lambda \underbrace{U^T x}_{\hat{x}} = \Lambda\hat{x} = \begin{bmatrix} \lambda_1\hat{x}_1 \\ \lambda_2\hat{x}_2 \\ \vdots \\ \lambda_n\hat{x}_n \end{bmatrix}$$

complicated matrix-vector product → scaling!

- We see that left-multiplying matrix $A$ in the original space is equivalent to left-multiplying the diagonal matrix $\Lambda$ w.r.t the new basis, which is merely scaling each coordinate by the corresponding eigenvalue.

# “Diagonalizing” matrix-vector multiplication

Taking a power of a matrix

Under the new basis, multiplying a matrix multiple times becomes much simpler as well. For example, suppose $\boxed{q = AAAx.}$

$$\hat{q} = U^T q = U^T AAAx = U^T U \Lambda U^T U \Lambda U^T U \Lambda U^T x = \Lambda^3 \hat{x} = \begin{bmatrix} \lambda_1^3 \hat{x}_1 \\ \lambda_2^3 \hat{x}_2 \\ \vdots \\ \lambda_n^3 \hat{x}_n \end{bmatrix}$$

$$A^{200} x = (\lambda^{200}) x$$

# "Diagonalizing" quadratic form

As a directly corollary, the quadratic form $x^T Ax$ can also be simplified under the new basis

$$\boxed{x^T Ax} = x^T U\Lambda U^T x = \hat{x}^T \Lambda \hat{x} = \boxed{\sum_{i=1}^{n} \lambda_i \hat{x}_i^2}$$

(Recall that with the old representation, $x^T Ax = \sum_{i=1,j=1}^{n} x_i x_j A_{ij}$ involves a sum of $n^2$ terms instead of $n$ terms in the equation above.)

# The definiteness of the matrix $A$ depends entirely on the sign of its eigenvalues

1. If $\boxed{\text{all } \lambda_i > 0}$, then the matrix $A$ is positive definite because $x^T Ax = \sum_{i=1}^n \lambda_i \hat{x}_i^2 > 0$ for any $\hat{x} \neq 0$.[1]
2. If all $\lambda_i \geq 0$, it is positive semidefinite because $x^T Ax = \sum_{i=1}^n \lambda_i \hat{x}_i^2 \geq 0$ for all $\hat{x}$.
3. Likewise, if all $\lambda_i < 0$ or $\lambda_i \leq 0$, then $A$ is negative definite or negative semidefinite respectively.
4. Finally, if $A$ has both positive and negative eigenvalues, say $\lambda_i > 0$ and $\lambda_j < 0$, then it is indefinite. This is because if we let $\hat{x}$ satisfy $\hat{x}_i = 1$ and $\hat{x}_k = 0, \forall k \neq i$, then $x^T Ax = \sum_{i=1}^n \lambda_i \hat{x}_i^2 > 0$. Similarly we can let $\hat{x}$ satisfy $\hat{x}_j = 1$ and $\hat{x}_k = 0, \forall k \neq j$, then $x^T Ax = \sum_{i=1}^n \lambda_i \hat{x}_i^2 < 0$.

[1] Note that $\hat{x} \neq 0 \Leftrightarrow x \neq 0$.

# Outline

1 Basic Concepts and Notation

2 Matrix Multiplication

3 Operations and Properties

4 Matrix Calculus

# Matrix Calculus

# The Gradient

Analogus to first derivative

vector/matrix scalar

Suppose that $f : \mathbb{R}^{m\times n} \to \mathbb{R}$ is a function that takes as input a matrix $A$ of size $m \times n$ and returns a real value. Then the ***gradient*** of $f$ (with respect to $A \in \mathbb{R}^{m\times n}$) is the matrix of partial derivatives, defined as:

$\nabla_A$

$A : (m \times n)$

$\nabla_A f : (m \times n)$

$$\nabla_A f(A) \in \mathbb{R}^{m\times n} = \begin{bmatrix} \frac{\partial f(A)}{\partial A_{11}} & \frac{\partial f(A)}{\partial A_{12}} & \cdots & \frac{\partial f(A)}{\partial A_{1n}} \\ \frac{\partial f(A)}{\partial A_{21}} & \frac{\partial f(A)}{\partial A_{22}} & \cdots & \frac{\partial f(A)}{\partial A_{2n}} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial f(A)}{\partial A_{m1}} & \frac{\partial f(A)}{\partial A_{m2}} & \cdots & \frac{\partial f(A)}{\partial A_{mn}} \end{bmatrix}$$

$(\nabla_A f)_{ij} = \frac{\partial f}{\partial A_{ij}}$

i.e., an $m \times n$ matrix with

$$(\nabla_A f(A))_{ij} = \frac{\partial f(A)}{\partial A_{ij}}.$$

# The Gradient

Note that the size of $\nabla_A f(A)$ is always the same as the size of $A$. So if, in particular, $A$ is just a vector $x \in \mathbb{R}^n$,

$$\underbrace{\nabla_x}_{(n\times 1)} f(x) = \underbrace{\begin{bmatrix} \frac{\partial f(x)}{\partial x_1} \\ \frac{\partial f(x)}{\partial x_2} \\ \vdots \\ \frac{\partial f(x)}{\partial x_n} \end{bmatrix}}_{(n\times 1)}. \qquad [\nabla_x]_i = \frac{\partial f}{\partial x_i}$$

$f: \mathbb{R}^n \to \mathbb{R}$

# The Gradient

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$$\nabla_x f(x) = \begin{bmatrix} \frac{\partial f(x)}{\partial x_1} \\ \frac{\partial f(x)}{\partial x_2} \\ \vdots \\ \frac{\partial f(x)}{\partial x_n} \end{bmatrix}.$$

It follows directly from the equivalent properties of partial derivatives that:

- $\nabla_x(f(x) + g(x)) = \nabla_x f(x) + \nabla_x g(x)$.
- For $t \in \mathbb{R}$, $\nabla_x(t\ f(x)) = t\nabla_x f(x)$.

Prove!

# The Hessian

— Analogous to 2nd derivative

(NOT matrix) vector scalar

Suppose that $f : \mathbb{R}^n \to \mathbb{R}$ is a function that takes a vector in $\mathbb{R}^n$ and returns a real number. Then the ***Hessian*** matrix with respect to $x$, written $\nabla_x^2 f(x)$ or simply as $H$ is the $n \times n$ matrix of partial derivatives,

$x : n \times 1$

$H_x : n \times n$

$f : \mathbb{R}^n \to \mathbb{R}$

$$\nabla_x^2 f(x) \in \mathbb{R}^{n\times n} = \begin{bmatrix} \frac{\partial^2 f(x)}{\partial x_1^2} & \frac{\partial^2 f(x)}{\partial x_1 \partial x_2} & \cdots & \frac{\partial^2 f(x)}{\partial x_1 \partial x_n} \\ \frac{\partial^2 f(x)}{\partial x_2 \partial x_1} & \frac{\partial^2 f(x)}{\partial x_2^2} & \cdots & \frac{\partial^2 f(x)}{\partial x_2 \partial x_n} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial^2 f(x)}{\partial x_n \partial x_1} & \frac{\partial^2 f(x)}{\partial x_n \partial x_2} & \cdots & \frac{\partial^2 f(x)}{\partial x_n^2} \end{bmatrix}.$$

$H_{ij} = \frac{\partial^2 f}{\partial x_i \, \partial x_j}$

In other words, $\nabla_x^2 f(x) \in \mathbb{R}^{n\times n}$, with

$$\boxed{(\nabla_x^2 f(x))_{ij} = \frac{\partial^2 f(x)}{\partial x_i \partial x_j}.}$$

# The Hessian

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Jacobian is different (if time permits)

Note that the Hessian is always symmetric, since

$$\frac{\partial^2 f(x)}{\partial x_i \partial x_j} = \frac{\partial^2 f(x)}{\partial x_j \partial x_i}.$$

## Gradients of Linear Functions

$f(x) = \sum b_i x_i$

$b : n \times 1$

$x : n \times 1$

$f : \mathbb{R}^n \to \mathbb{R}$

For $x \in \mathbb{R}^n$, let $f(x) = b^T x$ for some known vector $b \in \mathbb{R}^n$. Then

$$f(x) = \sum_{i=1}^{n} b_i x_i$$

$$[\nabla_x f]_i = \frac{\partial f}{\partial x_i} = b_i$$

so

$$\frac{\partial f(x)}{\partial x_k} = \frac{\partial}{\partial x_k} \sum_{i=1}^{n} b_i x_i = b_k.$$

From this we can easily see that $\boxed{\nabla_x b^T x = b}$. This should be compared to the analogous situation in single variable calculus, where $\partial/(\partial x)\ ax = a$.

# Gradients of Quadratic Function

$f: \mathbb{R}^n \to \mathbb{R}$

Now consider the quadratic function $f(x) = x^T Ax$ for $A \in \mathbb{S}^n$. Remember that

$$f(x) = \sum_{i=1}^{n} \sum_{j=1}^{n} A_{ij} x_i x_j .$$

$$[\nabla_x f]_i = \frac{\partial f}{\partial x_i}$$

To take the partial derivative, we'll consider the terms including $x_k$ and $x_k^2$ factors separately:

$$\frac{\partial f(x)}{\partial x_k} = \frac{\partial}{\partial x_k} \sum_{i=1}^{n} \sum_{j=1}^{n} A_{ij} x_i x_j$$

# Gradients of Quadratic Function

Now consider the quadratic function $f(x) = x^T Ax$ for $A \in \mathbb{S}^n$. Remember that

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To take the partial derivative, we'll consider the terms including $x_k$ and $x_k^2$ factors separately:

$$\begin{aligned} \frac{\partial f(x)}{\partial x_k} &= \frac{\partial}{\partial x_k} \sum_{i=1}^{n} \sum_{j=1}^{n} A_{ij} x_i x_j \\ &= \frac{\partial}{\partial x_k} \left[ \sum_{i \neq k} \sum_{j \neq k} A_{ij} x_i x_j + \sum_{i \neq k} A_{ik} x_i x_k + \sum_{j \neq k} A_{kj} x_k x_j + A_{kk} x_k^2 \right] \end{aligned}$$

# Gradients of Quadratic Function

Now consider the quadratic function $f(x) = x^T Ax$ for $A \in \mathbb{S}^n$. Remember that

$$f(x) = \sum_{i=1}^{n}\sum_{j=1}^{n} A_{ij}x_ix_j.$$

To take the partial derivative, we'll consider the terms including $x_k$ and $x_k^2$ factors separately:

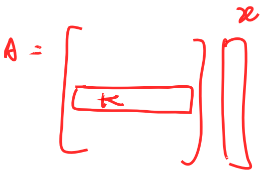
$$\begin{aligned}
\frac{\partial f(x)}{\partial x_k} &= \frac{\partial}{\partial x_k}\sum_{i=1}^{n}\sum_{j=1}^{n} A_{ij}x_ix_j \\
&= \frac{\partial}{\partial x_k}\left[\cancel{\sum_{i\neq k}\sum_{j\neq k}} A_{ij}x_ix_j + \sum_{i\neq k} A_{ik}x_ix_k + \sum_{j\neq k} A_{kj}x_kx_j + A_{kk}x_k^2\right] \\
&= \sum_{i\neq k} A_{ik}x_i + \sum_{j\neq k} A_{kj}x_j + 2A_{kk}x_k
\end{aligned}$$

# Gradients of Quadratic Function

Now consider the quadratic function $f(x) = x^T Ax$ for $A \in \mathbb{S}^n$. Remember that

$$f(x) = \sum_{i=1}^{n} \sum_{j=1}^{n} A_{ij} x_i x_j .$$

To take the partial derivative, we'll consider the terms including $x_k$ and $x_k^2$ factors separately:




$$\begin{aligned}
\frac{\partial f(x)}{\partial x_k} &= \frac{\partial}{\partial x_k} \sum_{i=1}^{n} \sum_{j=1}^{n} A_{ij} x_i x_j \\
&= \sum_{i \neq k} A_{ik} x_i + \sum_{j \neq k} A_{kj} x_j + 2A_{kk} x_k \\
&= \sum_{i=1}^{n} A_{ik} x_i + \sum_{j=1}^{n} A_{kj} x_j = \boxed{2 \sum_{i=1}^{n} A_{ki} x_i ,}
\end{aligned}$$

$\therefore \boxed{\nabla_x f = 2Ax}$

$\because$ A is symmetric

# Hessian of Quadratic Functions

Finally, let's look at the Hessian of the quadratic function $f(x) = x^T Ax$
In this case,

$$\frac{\partial^2 f(x)}{\partial x_k \partial x_\ell} = \frac{\partial}{\partial x_k}\left[\frac{\partial f(x)}{\partial x_\ell}\right] = \frac{\partial}{\partial x_k}\left[2\sum_{i=1}^{n} A_{\ell i} x_i\right] = 2A_{\ell k} = 2A_{k\ell}.$$

Therefore, it should be clear that $\boxed{\nabla_x^2 x^T Ax = 2A}$, which should be entirely expected (and again analogous to the single-variable fact that $\partial^2/(\partial x^2)\ ax^2 = 2a$).

# Recap

- $\nabla_x b^T x = b \quad = (b^T)^T$
- $\nabla_x^2 b^T x = 0$
- $\nabla_x x^T A x = 2Ax$ (if $A$ symmetric)
- $\nabla_x^2 x^T A x = 2A$ (if $A$ symmetric)

# Matrix Calculus Example: Least Squares

- Given a full rank matrix $A \in \mathbb{R}^{m\times n}$, and a vector $b \in \mathbb{R}^{m}$ such that $b \notin \mathcal{R}(A)$, we want to find a vector $x$ such that $Ax$ is as close as possible to $b$, as measured by the square of the Euclidean norm $\|Ax - b\|_2^2$. $(\ell_2)$

$$\arg\min_{x} \|Ax - b\|_2^2$$

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- Using the fact that $\|x\|_2^2 = x^T x$, we have

$$\|Ax - b\|_2^2 = (Ax - b)^T (Ax - b) = x^T A^T Ax - 2b^T Ax + b^T b$$

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- Using the fact that $\|x\|_2^2 = x^T x$, we have

$$\|Ax - b\|_2^2 = (Ax - b)^T (Ax - b) = x^T A^T A x - 2b^T A x + b^T b$$

- Taking the gradient with respect to $x$ we have: To minimize, set $\nabla = 0$

$$\begin{aligned} \nabla_x (x^T A^T A x - 2b^T A x + b^T b) &= \nabla_x x^T (A^T A) x - \nabla_x 2b^T A x + \cancel{\nabla_x b^T b} \\ &= 2A^T A x - 2A^T b \end{aligned}$$

$0$

$\nabla_x (2b^T A) x = (2b^T A)^T$

$= 2A^T b$

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- Taking the gradient with respect to $x$ we have:

$$\begin{aligned}\nabla_x (x^T A^T A x - 2b^T A x + b^T b) &= \nabla_x x^T A^T A x - \nabla_x 2b^T A x + \nabla_x b^T b \\ &= 2A^T A x - 2A^T b \quad = 0\end{aligned}$$

- Setting this last expression equal to zero and solving for $x$ gives the normal equations

normal eqn.s in class

$$\boxed{x = (A^T A)^{-1} A^T b}$$